J. David Muñoz-Bolaños

R&D Optical Engineer | Optical metrology specialist

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PROFESSIONAL SUMMARY

R&D Optical engineer based in **Innsbruck** with a magic combination of **Industrial Machine Vision** expertise and **High-Precision Optical Metrology**. Proven track record in developing illumination concepts, algorithmics (HALCON/Python), and hardware integration for automated inspection systems (LabVIEW/FPGA).

Experience bridging the gap between "code" and "hardware," from implementing inspection systems on high-speed industrial lines to correcting optical aberrations using MEMS devices. Eager to apply expertise in **optics, vision, and algorithms** to Besi's next-generation systems.

TECHNICAL COMPETENCIES

Vision Technology & Algorithmics
MVTec HALCON (Industrial software), optical quality control implementation, Image Processing Pipelines, Python, GPU Acceleration (CuPy) for real-time processing, Laser Profilometry, Camera/Laser Calibration.

Optics & Lab Management
Illumination/optical Design, Lab Maintenance & Safety, Adaptive Optics (SLM, Texas Instruments DMD), Interferometry, Beam Shaping, System Integration, Lithography Concepts.

Hardware Control
FPGA Real-time control (LabVIEW), Motion Control, High-speed Detectors (PMT/APD), Semiconductor-based MEMS integration.

Languages
English (Proficient), German (Intermediate/Working Proficiency), Spanish (Native).

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck — Innsbruck, Austria
PhD Researcher - Optical Systems Engineering — Jan 2022 – Present

- **High-Speed Hardware Integration (DMD):** Designed and built a control system for a **Texas Instruments DLP/DMD** to correct optical aberrations. This work mirrors die-attach alignment tasks, requiring high-speed/precision actuation and real-time feedback loops.
- **Algorithmics & GPU Acceleration:** Developed GPU-accelerated algorithms (Python/CuPy) to restore imaging quality in complex media. Optimized processing pipelines to handle large image stacks efficiently.
- **Optics Lab Management:** Responsible for the maintenance and improvement of the microscopy lab. Tasks include using laser safety protocols, handling fine optical components, and aligning complex beam paths for microscopic inspection.

- **Precision Metrology:** Implemented phase-stepping interferometry for nanometer-precision sensing, ensuring the highest optical restoration.

Leibniz Institute of Photonic Technology (IPHT) Jena, Germany
R&D Assistant Researcher Jun 2020 – Dec 2021

- **Prototyping:** Assembled multimodal imaging setups (Raman/Photothermal) from concept to functional prototype.
- **Hardware Integration:** Collaborated with scientific partners to qualify new detection methods for environmental microplastic analysis.

INTECOL S.A.S. Medellin, Colombia
Machine Vision Engineer Mar 2018 – Mar 2019

- **Industrial Deployment (OCR):** Led the full development cycle of a quality control system for bottling lines. Designed custom **LED illumination** concepts and programmed **HALCON** a lgorithms to reliably read and verify OCR codes on printed labels, achieving high traceability.
- **Safety-Critical Inspection:** Developed a **laser profilometry system** for the Metro de Medellin cable cars. Created the concept, integrated the vision hardware, and validated the system for measuring cable thickness in outdoor environments.
- **Production Transfer:** Managed on-site installation, calibration, and provided technical support and training for operators.

EDUCATION

Medical University of Innsbruck Austria
Ph.D. Advanced Microscopy & Optics Working available from March, 2026

Friedrich Schiller University Jena Germany
M.Sc. in Photonics 2019 – 2021

University of Cauca Colombia
B.Sc. in Physics Engineering 2012 – 2018

SELECTED PUBLICATIONS & AWARDS

- **J. D. Muñoz-Bolaños**, et al. "Scatter compensation for enabling deep two-photon microscopy using a MEMS phase light modulator," *Proc. SPIE* (2026).
- **J. D. Muñoz-Bolaños**, et al. "Wavefront corrections over large fields of view via beam cone tomography." *Proc. SPIE*, (2025).
- **Awards:** ZEISS Summer School (2025), COLFUTURO Grant (2019).

REFERENCES

Prof. Monika Ritsch-Marte
Medical University of Innsbruck
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Prof. Jürgen Popp
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